

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 1: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

Show ip route ospf

Part 2: Configure OSPF Passive Interfaces

a.

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? 192.168.13.1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 110

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

R2 don't send hello packets out. The neighbor's table don't show up then.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

En config t router ospf 1 no passive-default
s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

110 ~~110~~ 65 0

Is R2 listed as an OSPF neighbor to R3? Yes 2

Part 3: Change OSPF Metrics

Step 2: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

It may make the route change the cost higher or lower. Backup the route that do ~~not~~ want used unless primary goes down.

Step 3: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to 192.168.3.0/24 : R1 S0/0/1 + R3 G0/0 (781+1 = 782).

The cost to 192.168.23.0/30 : R1S0/0/1and R3 S0/0/1 (781+ 64 = 845).

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562.

$781 * 2 = 1562$.

Each serial link has 781 cost. The route to the 192.168.23.0/24 travels over two serial links.

Step 4: Change the route cost.

- c.Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF choose the route with the least cost. The least cost is R1-S0/0/0, R2-S0/0/1 and R3-G0/0, it's $781+781+1 = 1563$ costs. It is smaller than the R1-S0/0/1 + R3-G0/0, $1565+1 = 1566$.

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Router ID assignments determine the designated router and the backup designated router. If the router ID is tied to an

active interface, it may change when that interface goes down. Therefore, it should be set using the IP address of a

loopback interface.

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1. Why is the DR/BDR election process not a concern in this lab?

DR/BDR election process only occurs on multi-access networks. The serial links in the example

above are point-to-point, so no election takes place.

4

1. Why would you want to set an OSPF interface to passive?

Configuring a LAN interface as passive suppresses unnecessary OSPF routing information. This prevents the establishment of neighbor adjacencies on that link,

which conserves bandwidth.

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2. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF
Area _____

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3. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

Router(config-if)#ip address 10.0.0.0
255.255.255.0 _____

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4. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

RIP, OSPF and
EIGRP. _____

~~X~~ EIGRP 0